Module 8

Options – Solutions, Greeks and Volatility

October 20 & 22, 2025

BSM Equation for Dividend Paying Stock

We derive the pricing equation for the case S paying continuous dividend yield. This also lets us revisit the idea behind derivation of BSM equation.

Let f = derivatives instrument matures at T, S = share price, m = # of shares, r = continuous compounded risk free rate, q = continuous dividend yield. Consider again a hedged portfolio $\Pi = f - mS$. For $0 \le t < T$, under small time increment δt

$$\delta\Pi = \left(\frac{\partial f}{\partial t} + \mu S \frac{\partial f}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 f}{\partial S^2} \right)\delta t + \sigma S \frac{\partial f}{\partial S}\delta W_t - m\left((\mu + q)S\delta t + \sigma S \delta W_t \right)$$

The $qS\delta t$ term in δS is the capital gain from dividend payment during δt

Again choose $m = \dfrac{\partial f}{\partial S}$,

$$\delta\Pi = \left(\frac{\partial f}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 f}{\partial S^2} \right)\delta t - qS\frac{\partial f}{\partial S}\delta t$$

$$\rightarrow \quad \delta\Pi = \left(\frac{\partial f}{\partial t} - qS\frac{\partial f}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 f}{\partial S^2} \right)\delta t = r\Pi\delta t = r\left(f - S\frac{\partial f}{\partial S} \right)\delta t$$

$$\rightarrow \quad \frac{\partial f}{\partial t} + (r-q)S\frac{\partial f}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 f}{\partial S^2} - rf = 0$$

This is the BSM equation for a derivative f on a dividend paying stock S

Solving the BSM PDE (*):

Find f that satisfies:

- BSM Equation
- final condition at T that gives the desired option payoff

Simplifying the equation:

- $f(S,t) = e^{-r(T-t)} g(S,t)$

  → $\frac{\partial g}{\partial t} + (r-q)S\frac{\partial g}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 g}{\partial S^2} = 0$

  → get rid of the rf term

- $Y = \ln(S)$

  → $\frac{\partial g}{\partial t} + (r - q - \frac{\sigma^2}{2})\frac{\partial g}{\partial Y} + \frac{1}{2}\sigma^2 \frac{\partial^2 g}{\partial Y^2} = 0$,

  ($g(S,t)$ → $g(e^Y,t)$)

  → constant coefficients

- $\tau = T - t$

  → $\frac{\partial g}{\partial \tau} = (r - q - \frac{\sigma^2}{2})\frac{\partial g}{\partial Y} + \frac{1}{2}\sigma^2 \frac{\partial^2 g}{\partial Y^2}$,

  ($g(e^Y,t)$ → $g(e^Y, T-\tau)$)

  → final condition → initial condition

- $X = Y + (r - q - \frac{\sigma^2}{2})\tau$

  → $\frac{\partial g}{\partial \tau} = \frac{1}{2}\sigma^2 \frac{\partial^2 g}{\partial X^2}$,

  ($g(e^Y, T-\tau)$ → $g(e^{X-(r-q-\frac{\sigma^2}{2})\tau}, T-\tau)$)

  → get rid of the first derivative term

$h(X,\tau) \equiv g(e^{X-(r-q-\frac{\sigma^2}{2})\tau}, T-\tau)$

→ $\begin{cases} \frac{\partial h}{\partial \tau} = \frac{1}{2}\sigma^2 \frac{\partial^2 h}{\partial X^2}, & -\infty < X < \infty, \tau \geq 0 \\ h(X,0) = h_0(X) & -\infty < X < \infty \end{cases}$

→ "heat equation" in physics

- Call option: $h_0(X) = \max(e^X - K, 0)$
- Put option: $h_0(X) = \max(K - e^X, 0)$

In physics heat equations are often solved via Fourier transform (with appropriate boundary conditions)

Here we do something different:

$$h(X,\tau) = \frac{u(\eta)}{\sqrt{\tau}} \ , \ \eta = \frac{X - X'}{\sqrt{\tau}} \ , \ -\infty < \eta < \infty$$

$$\rightarrow \ \sigma^2 \frac{d^2u}{d\eta^2} + \frac{d(\eta u)}{d\eta} = 0$$

$$\rightarrow \ \sigma^2 \frac{du}{d\eta} + \eta u = c_0$$

Choose $c_0 = 0$ for a particular solution:

$$\rightarrow \ u = ce^{-\frac{\eta^2}{2\sigma^2}}$$

Normalize u such that $\int_{-\infty}^{\infty} u d\eta = 1$

$$\rightarrow \ u = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{\eta^2}{2\sigma^2}}$$

$$\rightarrow \ h(X,\tau;X') = \frac{1}{\sqrt{2\pi\tau}\sigma} e^{-\frac{(X-X')^2}{2\sigma^2\tau}} \ \text{(Green's function)}$$

The substitution $h(X,\tau) = \frac{u(\eta)}{\sqrt{\tau}}$, $\eta = \frac{X - X'}{\sqrt{\tau}}$ is a form of "similarity transformation" which tries to find solutions that depend on some groups of independent variables instead of each variable separately

Linearity → all h with different X' are also solutions to the heat equation

$$\rightarrow \ h(X,\tau) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\tau}\sigma} h_0(X') e^{-\frac{(X-X')^2}{2\sigma^2\tau}} dX' \ \text{is also solution}$$

$$\lim_{\tau \to 0} \frac{1}{\sqrt{2\pi\tau}\sigma} e^{-\frac{(X-X')^2}{2\sigma^2\tau}} = \delta(X - X') \ \text{(Dirac's } \delta\text{-function)}$$

$$\delta(X-X') = \begin{cases} 0 & X \neq X' \\ \int_{-\infty}^{\infty} \delta(X-X')dX' = 1 \end{cases}$$

$$\lim_{\tau \to 0} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\tau}\sigma} h_0(X') e^{-\frac{(X-X')^2}{2\sigma^2\tau}} dX' = h_0(X)$$

→ $h(X,\tau) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\tau}\sigma} h_0(X') e^{-\frac{(X-X')^2}{2\sigma^2\tau}} dX'$ is the solution of the heat equation with initial value $h_0(X)$

Option pricing:

Substitute

$$\tau = T - t, \quad X = \ln(S) + (r - q - \frac{\sigma^2}{2})(T-t)$$

for the expression of $f(S,t)$:

$$f(S,t) = e^{-r(T-t)} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\tau}\sigma} h_0(X') e^{-\frac{(X-X')^2}{2\sigma^2\tau}} dX' = e^{-r(T-t)} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi(T-t)}\sigma} f_T(e^{X'}) e^{-\frac{\left(\ln S + (r-q-\frac{\sigma^2}{2})(T-t) - X'\right)^2}{2\sigma^2(T-t)}} dX'$$

→ $f(S,t) = e^{-r(T-t)} \int_0^{\infty} \frac{1}{\sqrt{2\pi(T-t)}\sigma} f_T(S') e^{-\frac{\left(\ln S' - \ln S - (r-q-\frac{\sigma^2}{2})(T-t)\right)^2}{2\sigma^2(T-t)}} \frac{dS'}{S'}$

$\frac{1}{\sqrt{2\pi(T-t)}\sigma} \frac{1}{S'} e^{-\frac{\left(\ln S' - \ln S - (r-q-\frac{\sigma^2}{2})(T-t)\right)^2}{2\sigma^2(T-t)}}$: probability density for the lognormal distribution

→ "risk neutral probability density"

→ $f(S,t) = e^{-r(T-t)} E_t\left[f_T(S')\right]$

<u>Examples</u>:

- European call: $f_T(S) = \max(S-K, 0)$

$$c(S,t) = Se^{-q(T-t)}N(d_1) - Ke^{-r(T-t)}N(d_2)$$

- European put: $f_T(S) = \max(K - S, 0)$

$$p(S,t) = -Se^{-q(T-t)}N(-d_1) + Ke^{-r(T-t)}N(-d_2)$$

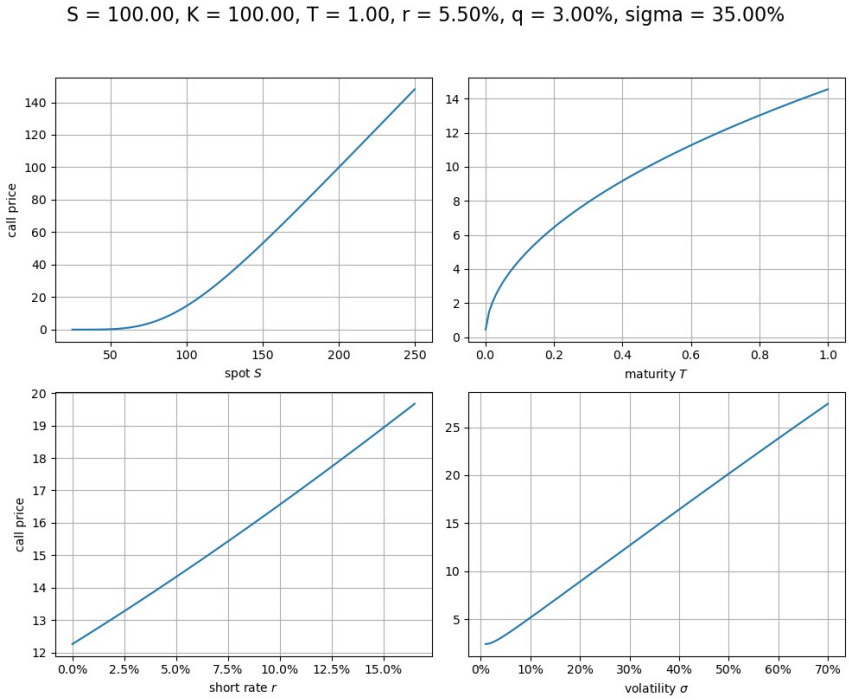
$$d_1 = \frac{\ln\left(\frac{S}{K}\right) + \left(r - q + \frac{\sigma^2}{2}\right)(T-t)}{\sigma\sqrt{T-t}}$$

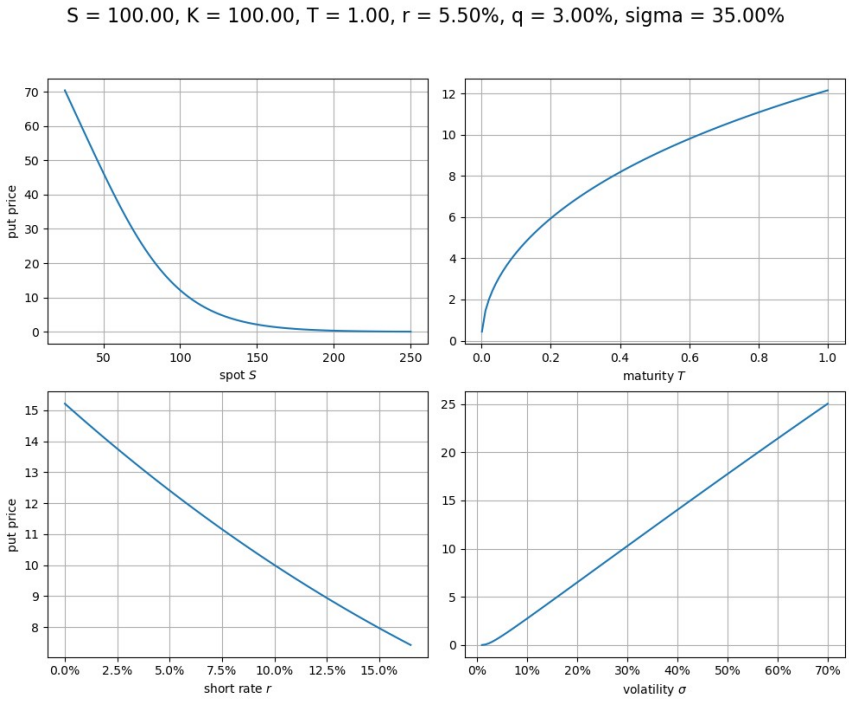
$$d_2 = \frac{\ln\left(\frac{S}{K}\right) + \left(r - q - \frac{\sigma^2}{2}\right)(T-t)}{\sigma\sqrt{T-t}} = d_1 - \sigma\sqrt{T-t}$$

Exercise:

Compute the expectation $E_t[.]$ for call/put options explicitly to derive the above formulas.

Numerical examples:





Greeks

Greeks are measure of sensitivity of option valuation wrt changes in pricing parameters:

- Delta: $\Delta = \dfrac{\partial f}{\partial S}$

  European call: $\Delta = e^{-q(T-t)}N(d_1)$

  European put: $\Delta = -e^{-q(T-t)}N(-d_1)$

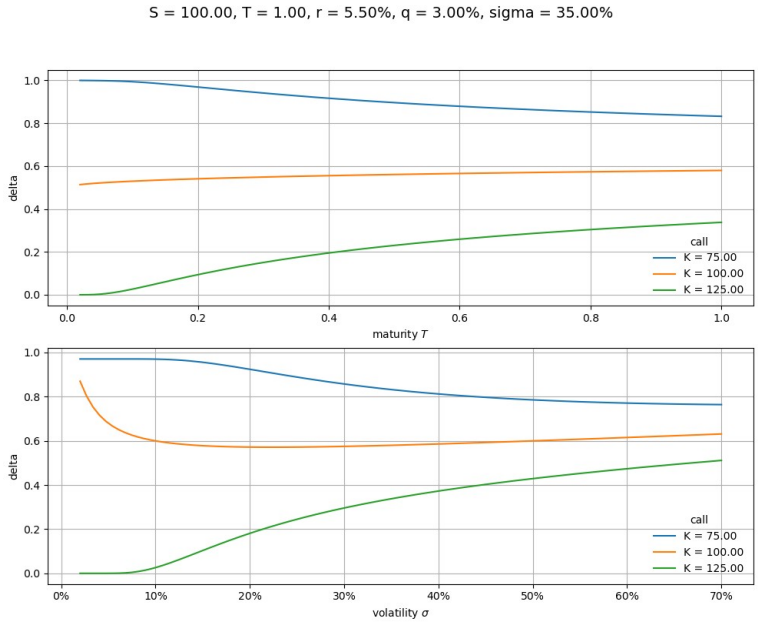
  - $c = \Delta S + B_t^c \ , \quad B_t^c = -Ke^{-r(T-t)}N(d_2)$
  - $p = \Delta S + B_t^p \ , \quad B_t^p = Ke^{-r(T-t)}N(-d_2)$

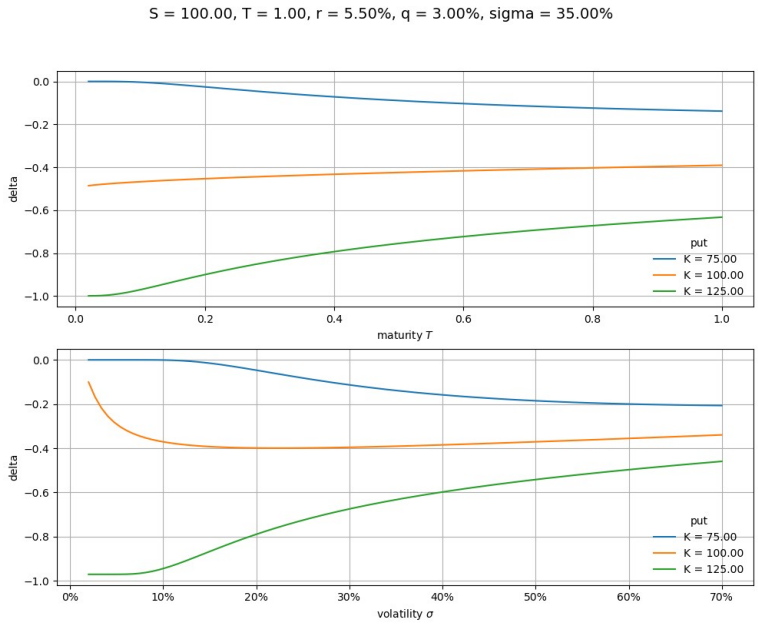
- Hedging option f with m shares:

  $\Pi = f - mS \ , \quad m = \dfrac{\partial f}{\partial S} = \Delta$

  → Π is instantaneous "delta neutral"

- Replicating options with shares and cash (zero coupon bonds)
  - amount of shares = option delta Δ
  - amount of cash = $\mp$ PV of option strike $\times$ probability of option ends ITM





- Gamma: $\Gamma = \frac{\partial^2 f}{\partial S^2}$

  European call: $\Gamma = \frac{e^{-q(T-t)} n(d_1)}{\sigma S \sqrt{T-t}}$

  European put: $\Gamma = \frac{e^{-q(T-t)} n(d_1)}{\sigma S \sqrt{T-t}}$

  $n(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$

  Gamma: rate of change of delta, convexity of the option position

  - Delta hedging in practice:
    - dynamic
    - discrete
    - can be biased

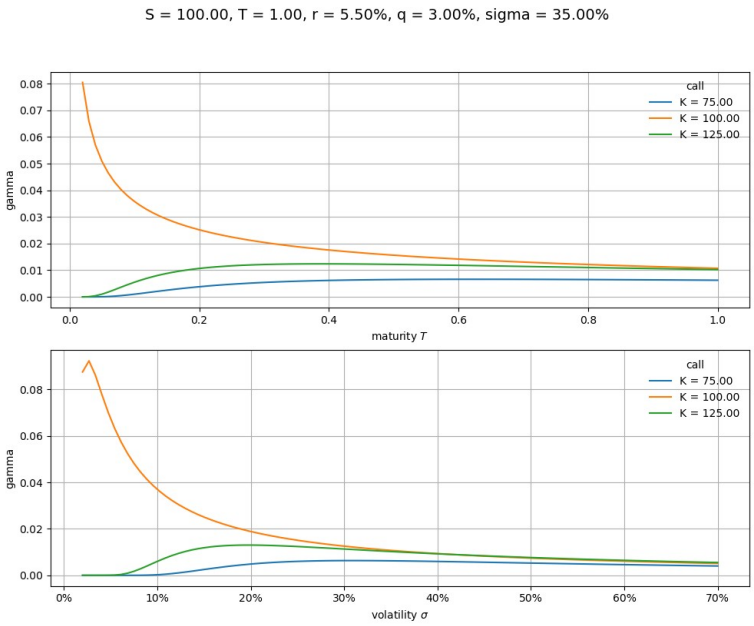
    Actual strategies depend on market friction for the underlying (transaction cost, borrow cost), nature of the position, etc

  - Monetizing option convexity:

    Trading position – high convexity, but not too close to maturity

    How can such a position be established? → buy "cheap" volatility

  - Effect of option hedging on cash markets:
    - Long gamma → volatility damping
    - Short gamma → volatility amplifying

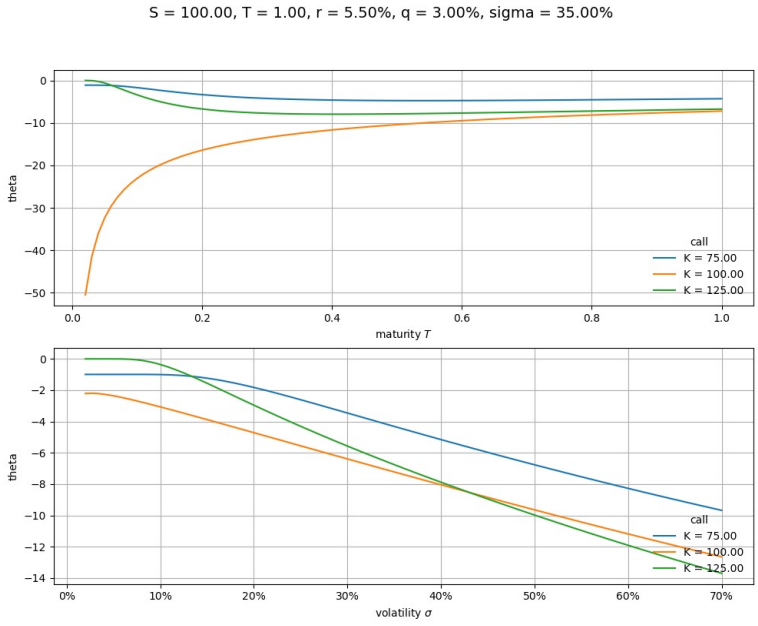


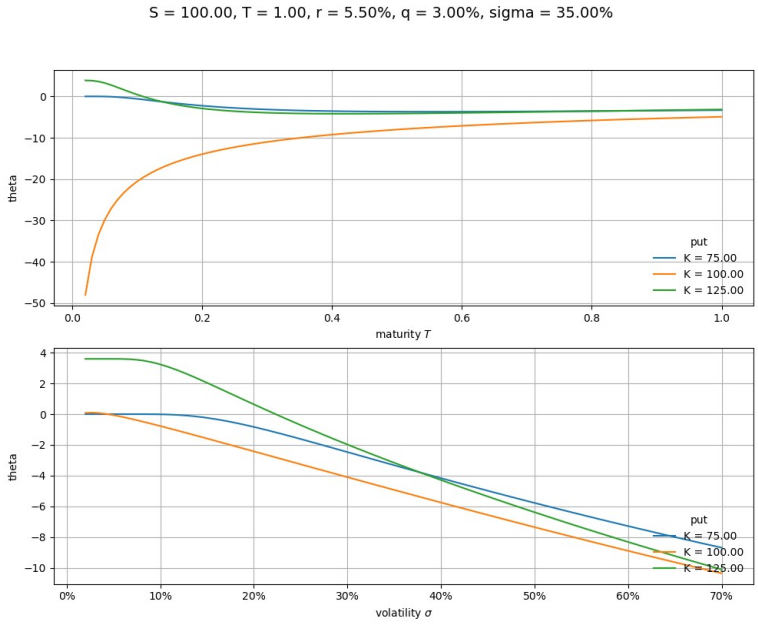
- Theta: $\Theta = \dfrac{\partial f}{\partial t}$

European call: $\Theta = -\dfrac{\sigma S e^{-q(T-t)} n(d_1)}{2\sqrt{T-t}} + qSe^{-q(T-t)}N(d_1) - rKe^{-r(T-t)}N(d_2)$

European put: $\Theta = -\dfrac{\sigma S e^{-q(T-t)} n(-d_1)}{2\sqrt{T-t}} - qSe^{-q(T-t)}N(-d_1) - rKe^{-r(T-t)}N(-d_2)$

Theta measures time decay of the option value

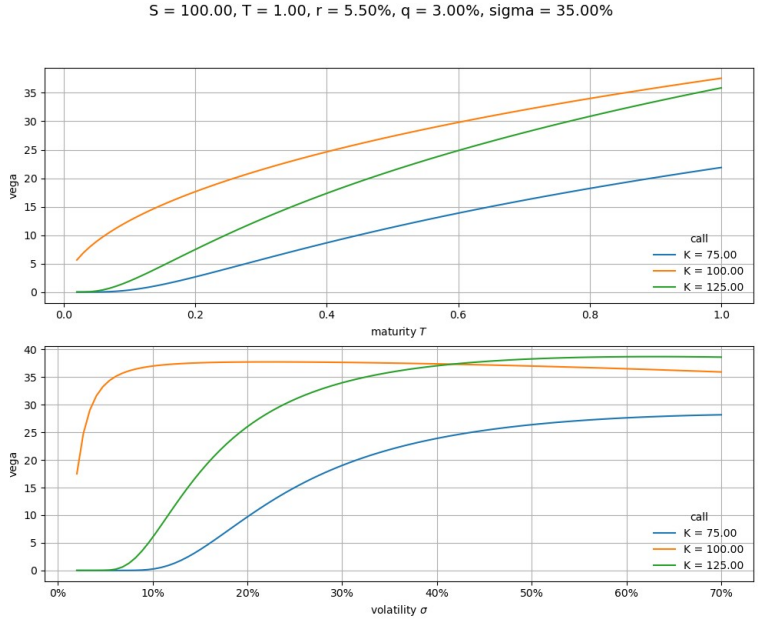




- Vega: $vega = \frac{\partial f}{\partial \sigma}$

  European call: $vega = S\sqrt{T-t}e^{-q(T-t)}n(d_1)$

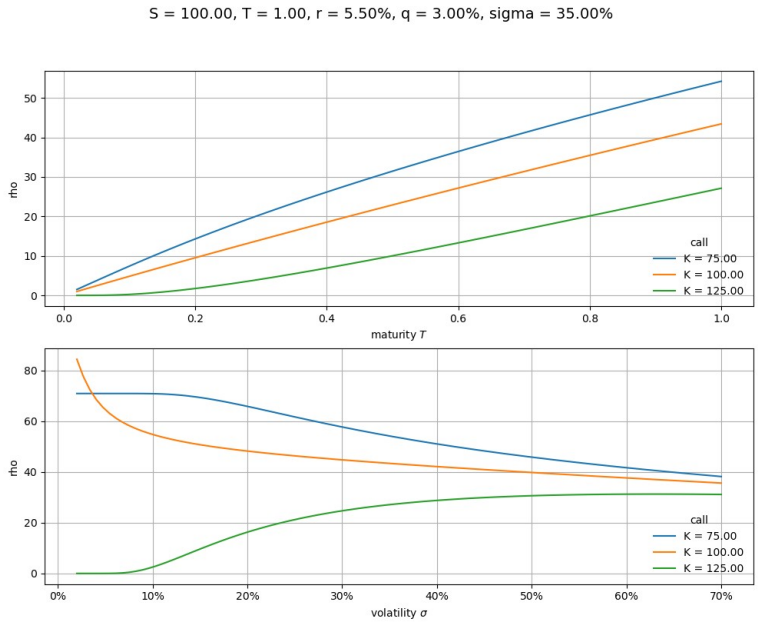
  European put: $vega = S\sqrt{T-t}e^{-q(T-t)}n(d_1)$

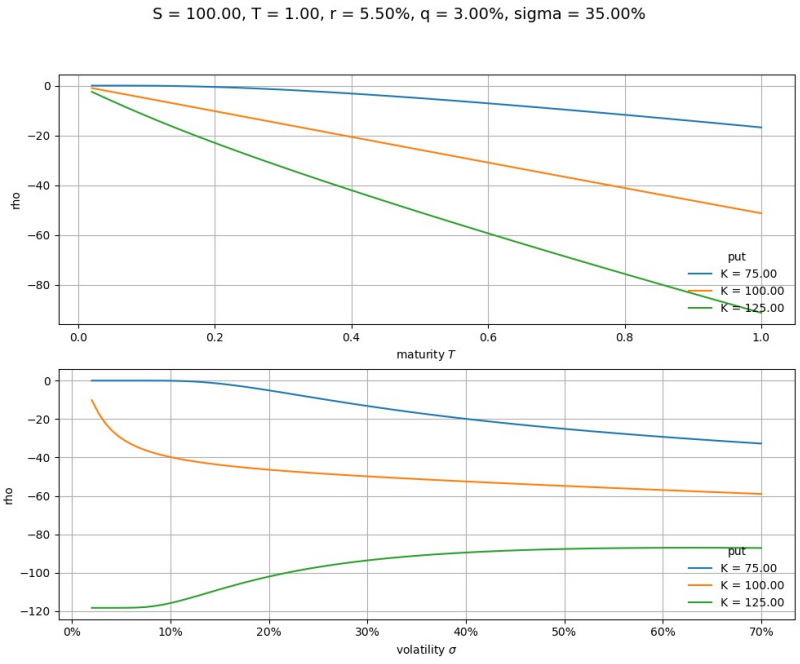


- Rho: $\rho = \frac{\partial f}{\partial r}$

European call: $\rho = K(T-t)e^{-r(T-t)}N(d_2)$

European put: $\rho = -K(T-t)e^{-rT-t)}N(-d_2)$





Exercises:

Verify the expressions for the greeks.

Relationship among greeks:

$$\frac{\partial f}{\partial t}+(r-q)S\frac{\partial f}{\partial S}+\frac{1}{2}\sigma^2S^2\frac{\partial^2 f}{\partial S^2}-rf=0 \quad \rightarrow \quad \Theta+(r-q)S\Delta+\frac{1}{2}\sigma^2S^2\Gamma-rf=0$$

For a delta neutral portfolio $\Pi=\sum_{i=1}^{N}w_i f_i$ (w_i can be either positive or negative),

$$\Delta_\Pi=\frac{\partial \Pi}{\partial S}=0 \quad \rightarrow \quad \Theta_\Pi+\frac{1}{2}\sigma^2S^2\Gamma_\Pi=r\Pi$$

Rule of thumb: large negative theta → large positive gamma (and vice versa)

Some other sensitivities:

- $\Phi=\frac{\partial f}{\partial q}$
- $vanna=\frac{\partial^2 f}{\partial S\partial\sigma}=\frac{\partial\Delta}{\partial\sigma}=\frac{\partial vega}{\partial S}$
- $volga=\frac{\partial^2 f}{\partial\sigma^2}=\frac{\partial vega}{\partial\sigma}$
- $charm=\frac{\partial^2 f}{\partial S\partial t}=\frac{\partial\Delta}{\partial t}=\frac{\partial\Theta}{\partial S}$

The most scrutinized EOD risk exposures from a derivative trading desk are $delta, $vega, $theta & $rho. Higher order sensitivities may also be examined for desks dealing with complex financial engineering products (see later in the course).

Implied Volatility

- What one needs for the BSM equation is the future realized volatility during the life of the option (t,T)

- Historical volatility may be used to help forecast future volatility (e.g., GARCH (Generalized AutoRegressive Conditional Heteroskedasticity))
- Assume that yield curve is stable and dividend forecast is accurate (reasonable to assume for short dated options), option prices itself may be used to extract information on future volatility

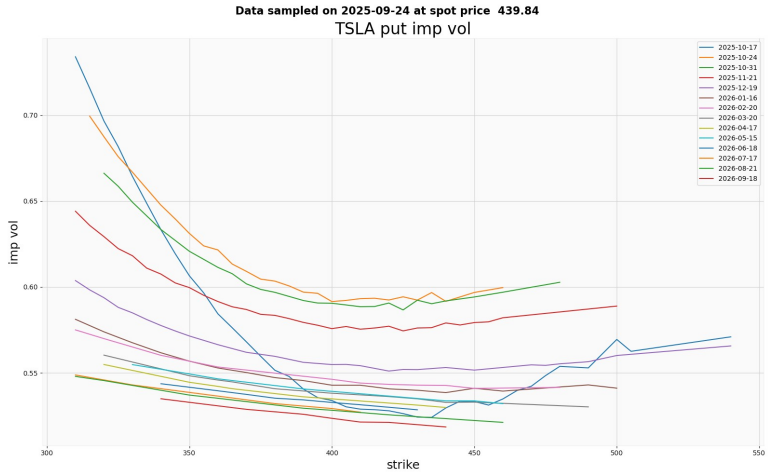
  Option price → (BSM Eqn) → option price implied volatility

  → implied volatility as option market expectation on future realized volatility
- Volatility term structure $\sigma(T)$:
  - Expected as different tenor may experience different degree of uncertainty
- Volatility smile (skew) $\sigma(K)$:
  - Prior to 1987 October "Black Monday": flat implied volatility
  - After Black Monday:

    Extra premium for downside risk → volatility skew

    (recall BSM model does not hedge jump risk)

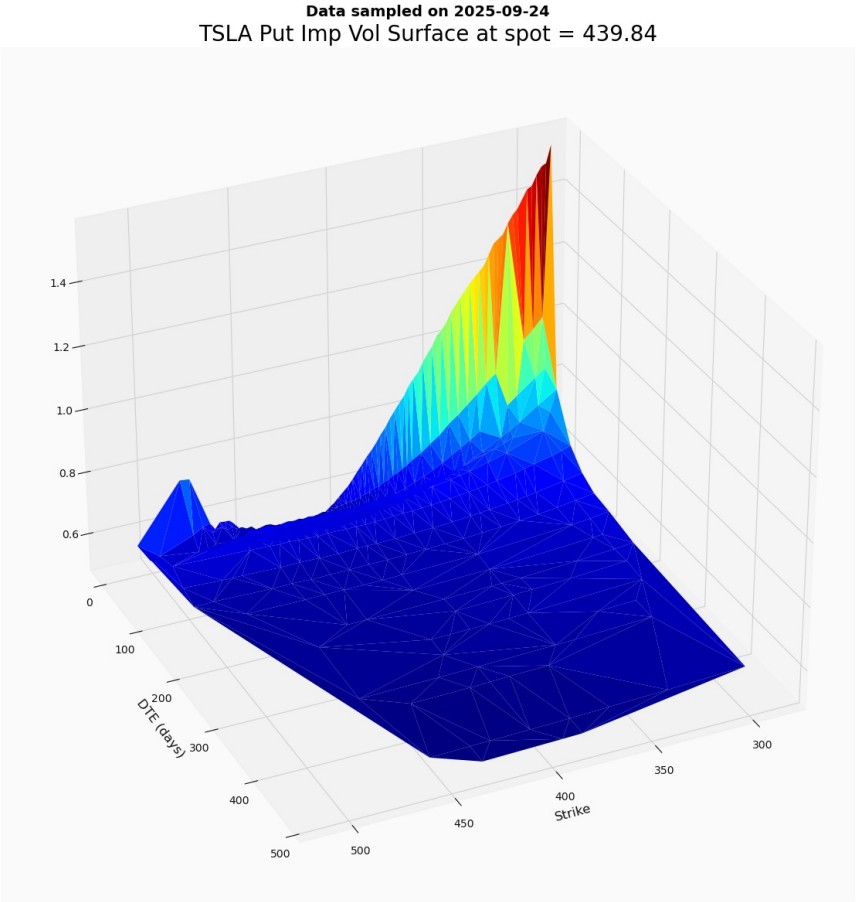
    Example: TSLA implied vols from the same data sampled in Module 5



  - Negative correlation between equity price and volatility
  - Volatility skew as an indication of inadequacy of BSM model

- Time and strike dependence of volatility → "volatility surface" $\sigma(T,K)$

  Example: TSLA implied vol surface from the same data sampled in Module 5



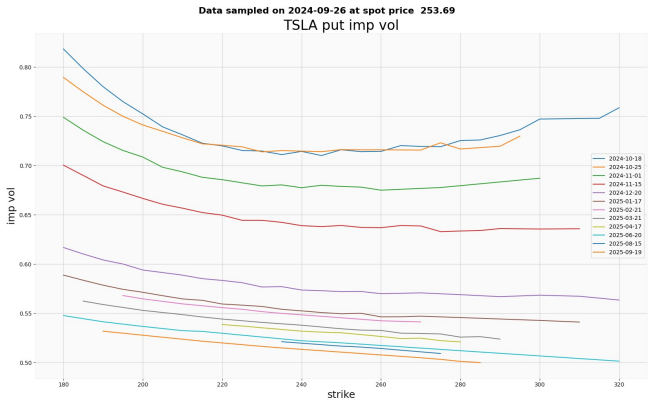
- Constraints on volatility surface (see the general properties of options discussed in Module 5):
  - Constraints on the degree of the vol skews

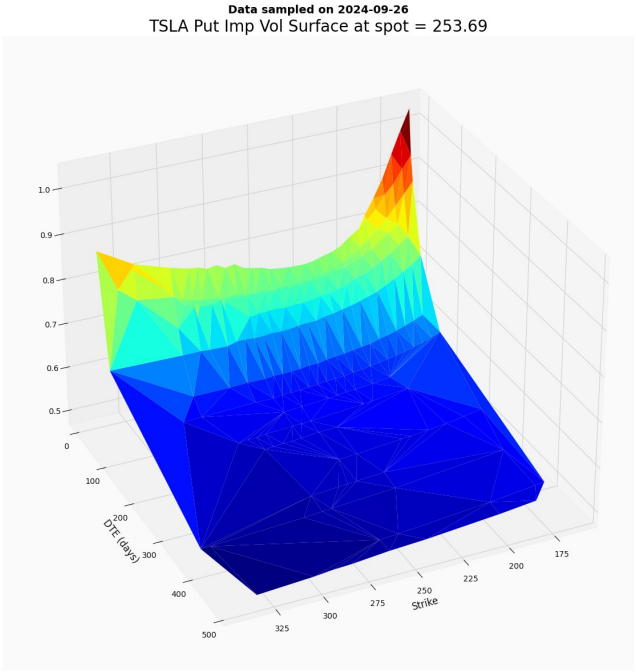
    → preventing vertical (i.e., call/put) spread arbitrage
  - Constraints on the degree of inversion of the vol term structures

    → preventing calendar spread arbitrage
- Option strategies depend on the shape of volatility surface:
  - Sell expensive vol, buy cheap vol
- Volatility dynamics:
  - Market expectation on future realized volatility changes over time
  - Risk premium for options at different strike changes over time

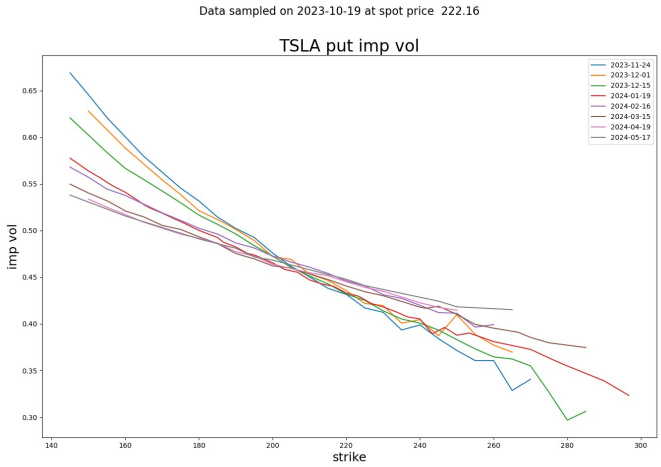
  → stochastic volatility (surface)

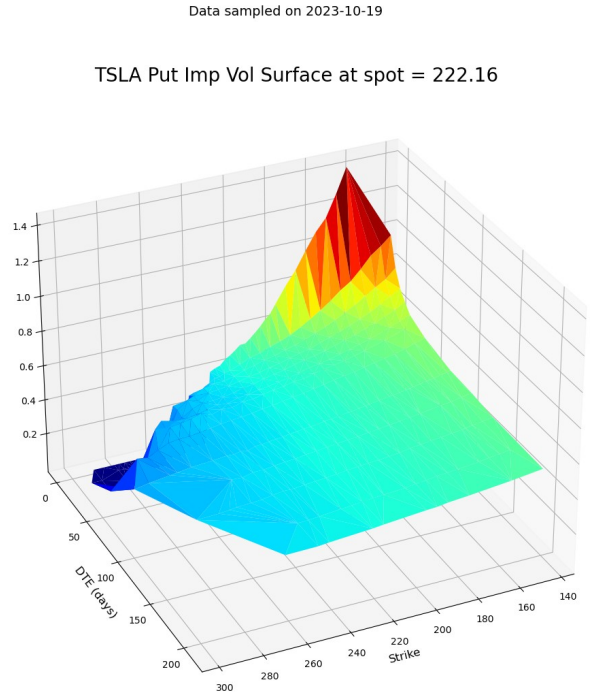
Example: TSLA implied vols sampled a year ago (2024-9-26):





Example: TSLA implied vols sampled 2 year ago (2023-10-19):





- P&L Decomposition:

  Consider a delta hedged portfolio $\Pi = f - \Delta S$. Its P&L in $(t, t+\delta t)$

  $$\Pi_{P\&L}(t, t+\delta t) = \delta f - \Delta \delta S - \Delta S q \delta t - (f - \Delta S) r \delta t$$

  where the last term is the finance cost for maintaining the portfolio. Taylor expansion for f :

$$\delta f = \Theta \delta t + \frac{1}{2}\Gamma \delta S^2 + \Delta \delta S + v\delta\sigma + ...$$

$$\Pi_{P\&L}(t, t+\delta t) = \Theta \delta t + \frac{1}{2}\Gamma \delta S^2 - \Delta S q \delta t - (f - \Delta S) r \delta t + v\delta\sigma + ...$$

$$= (\Theta + (r-q)\Delta S - rf)\delta t + \frac{1}{2}\Gamma \delta S^2 + v\delta\sigma + ...$$

$$= \frac{1}{2}\Gamma \delta S^2 - \frac{1}{2}\sigma^2 S^2 \Gamma + v\delta\sigma + ... \qquad (f \text{ satisfies BSM equation})$$

σ above is the option implied volatility and v is the option vega. For a small δt,

$\delta S^2 \approx \sigma_R^2 S^2 \delta t$

where σ_R is the realized volatility during $(t, t+\delta t)$. This leads to

$$\Pi_{P\&L}(t, t+\delta t) = \frac{1}{2}\Gamma S^2 \left(\sigma_R^2 - \sigma_I^2\right)\delta t + v\delta\sigma + ...$$

where the implied volatility σ_I is denoted explicitly.

- Gamma P&L: $\frac{1}{2}\Gamma S^2 \left(\sigma_R^2 - \sigma_I^2\right)\delta t$
- Vega P&L: $v\delta\sigma$

→ make money if one can "buy volatility cheap"

- Volatility as an asset class
  - Historical patterns of realized vol vs option implied vol
  - Covered call as a volatility strategy:

    Call buyers (speculators, hedgers (for the upside!)) may not be price sensitive → higher option premium than it "should"
  - Chicago Board Options Exchange (CBOE) Volatility Index – VIX, VIX futures, VIX options
  - Variance swaps:

    Payoff = realized variance in the specified period – strike variance

    Realized variance is usually computed from log returns

Option Pricing & Hedging via ML

- Solving American option problems via ML
- "Deep hedging"
- Model independent pricing and hedging

  → need for synthetic data generation (GANs, generative diffusion, …)
- …

(*) See "Paul Wilmott on Quantitative Finance", 2nd edition, Paul Wilmott, 2007